LW3: Merged Data, DAX, and DAX Function Insights

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Part 1: Review the Merged Dataset

OrderID	CustomerID	OrderDate	SalesAmount	CustomerName	City	Region
O003	C137	Tuesday, March 11, 2025	800	Customer 3	Butuan	Mindanao
O007	C091	Friday, February 7, 2025	1200	Customer 7	Zamboanga	Mindanao
O009	C027	Saturday, March 29, 2025	45000	Customer 9	Cagayan de Oro	Mindanao
O010	C094	Wednesday, February 19, 2025	1200	Customer 10	Cagayan de Oro	Mindanao
O012	C146	Saturday, March 1, 2025	3000	Customer 12	Cagayan de Oro	Mindanao
O013	C131	Friday, March 7, 2025	1200	Customer 13	Cagayan de Oro	Mindanao
O016	C173	Sunday, March 2, 2025	45000	Customer 16	Butuan	Mindanao
O024	C030	Monday, February 17, 2025	1200	Customer 24	Cagayan de Oro	Mindanao
O025	C100	Friday, March 28, 2025	2200	Customer 25	Davao	Mindanao
O028	C127	Friday, January 10, 2025	800	Customer 28	Cagayan de Oro	Mindanao
O030	C122	Tuesday, March 11, 2025	3000	Customer 30	Davao	Mindanao
O036	C077	Saturday, March 29, 2025	3000	Customer 36	Davao	Mindanao
O042	C132	Wednesday, March 19, 2025	3000	Customer 42	Zamboanga	Mindanao
O043	C020	Wednesday, February 5, 2025	3000	Customer 43	Cagayan de Oro	Mindanao
O044	C024	Tuesday, March 4, 2025	800	Customer 44	Zamboanga	Mindanao
O047	C046	Wednesday, January 22, 2025	800	Customer 47	Zamboanga	Mindanao
O049	C023	Wednesday, January 8, 2025	1200	Customer 49	Zamboanga	Mindanao
O054	C018	Tuesday, January 7, 2025	1500	Customer 54	Cagayan de Oro	Mindanao
O058	C162	Monday, March 24, 2025	7500	Customer 58	Butuan	Mindanao
O060	C174	Monday, March 10, 2025	45000	Customer 60	Butuan	Mindanao
O061	C184	Wednesday, March 19, 2025	2200	Customer 61	Cagayan de Oro	Mindanao
O062	C076	Monday, March 24, 2025	2200	Customer 62	Cagayan de Oro	Mindanao
O067	C087	Friday, March 28, 2025	7500	Customer 67	Davao	Mindanao
O075	C079	Wednesday, January 1, 2025	3000	Customer 75	Davao	Mindanao

Table: the Merged_Sales_Data (200 rows)

Data

Search

Customers

the Merged_Sales_Data

Guide Questions

1. Is the dataset in flat-table format?

The dataset uses a single-table structure where all records are organized in rows and columns. Each row contains the details of one order entry.

2. Which columns describe customers?

Customer-related information can be found in the CustomerID, CustomerName, City, and Region columns.

3. Which columns represent transactions?

The columns that contain transaction data are OrderID, OrderDate, and Sales Amount, which show the order details and the amount of the sale.

Part 2: Identify Fact and Dimension Tables

Create DimCustomer

Queries [4]

- the Merged_Sales_Data
- Customers
- DimCustomer
- FactSales

```
= Table.Distinct(#"Removed Columns", {"CustomerID"})
```

	CustomerID	CustomerName	City	Region
1	C009	Customer 1	Legazpi	Luzon
2	C156	Customer 2	Manila	Luzon
3	C137	Customer 3	Butuan	Mindanao
4	C127	Customer 4	Dumaguete	Visayas
5	C105	Customer 5	Dumaguete	Visayas
6	C054	Customer 6	Cebu	Visayas
7	C091	Customer 7	Zamboanga	Mindanao
8	C016	Customer 8	Baguio	Luzon
9	C027	Customer 9	Cagayan de Oro	Mindanao
10	C094	Customer 10	Cagayan de Oro	Mindanao
11	C146	Customer 12	Cagayan de Oro	Mindanao
12	C131	Customer 13	Cagayan de Oro	Mindanao
13	C170	Customer 14	Baguio	Luzon
14	C021	Customer 15	Tacloban	Visayas
15	C173	Customer 16	Butuan	Mindanao
16	C115	Customer 17	Baguio	Luzon
17	C192	Customer 18	Iloilo	Visayas
18	C159	Customer 19	Baguio	Luzon
19	C026	Customer 21	Manila	Luzon
20	C086	Customer 22	Legazpi	Luzon
21	C051	Customer 23	Tacloban	Visayas
22	C030	Customer 24	Cagayan de Oro	Mindanao
23	C100	Customer 25	Davao	Mindanao
24	C088	Customer 26	Cebu	Visayas

4 COLUMNS, 127 ROWS Column profiling based on top 1000 rows

Create FactSales

Queries [4]

- the Merged_Sales_Data
- Customers
- DimCustomer
- FactSales

```
= Table.RemoveColumns(Source,{"CustomerName", "City", "Region"})
```

	OrderID	CustomerID	OrderDate	SalesAmount
1	O001	C009	1/4/2025	3000
2	O002	C156	2/12/2025	7500
3	O003	C137	3/11/2025	800
4	O004	C127	1/1/2025	45000
5	O005	C105	3/3/2025	1200
6	O006	C054	3/29/2025	2200
7	O007	C091	2/7/2025	1200
8	O008	C016	1/17/2025	2200
9	O009	C027	3/29/2025	45000
10	O010	C094	2/19/2025	1200
11	O011	C094	2/22/2025	3000
12	O012	C146	3/1/2025	3000
13	O013	C131	3/7/2025	1200
14	O014	C170	2/12/2025	45000
15	O015	C021	1/6/2025	800
16	O016	C173	3/2/2025	45000
17	O017	C115	2/15/2025	45000
18	O018	C192	2/10/2025	800
19	O019	C159	2/5/2025	7500
20	O020	C156	1/28/2025	800
21	O021	C026	1/22/2025	800
22	O022	C086	1/14/2025	800
23	O023	C051	1/14/2025	3000
24	O024	C030	2/17/2025	1200

4 COLUMNS, 200 ROWS Column profiling based on top 1000 rows

Guide Questions

1. Why must dimension tables contain unique keys?

Dimension tables need unique keys to ensure that every record can be clearly distinguished. This allows the fact table to correctly connect to the right dimension record, maintaining accurate relationships between the tables.

2. What problems occur if duplicates exist in DimCustomer?

If duplicate values appear in Dim customer, it can disrupt the relationship with Factsales. This may lead to incorrect report outputs, such as duplicated calculations or sales totals being counted more than once for the same customer.

Part 3: Define Primary & Foreign Keys

Identify Keys

Table	Key	Type
DimCustomer	CustomerID	Primary Key
FactSales	CustomerID	Foreign Key
FactSales	OrderID	Primary Key

Guide Questions

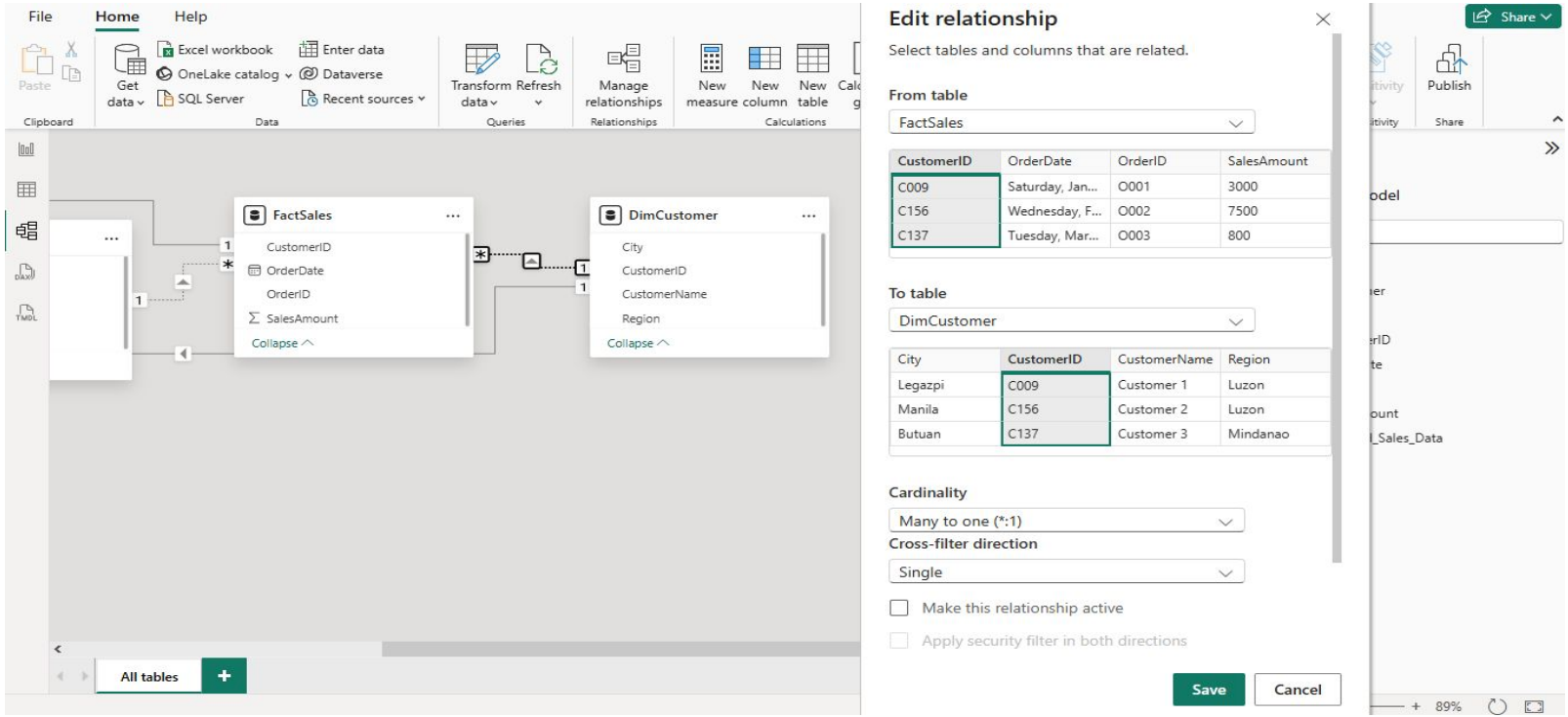
1. What is a primary key?

A primary key is a field in a table that ensures every row can be uniquely recognized. In the example, CustomerID functions as the primary key in DimCustomer, while OrderID serves as the primary key in FactSales, meaning each value in those columns must be unique.

2. Why is CustomerID a foreign key in FactSales?

CustomerID is a foreign key because it links the sales record in FactSales to the customer in DimCustomer.

Part 4: Create Relationships



Guide Questions

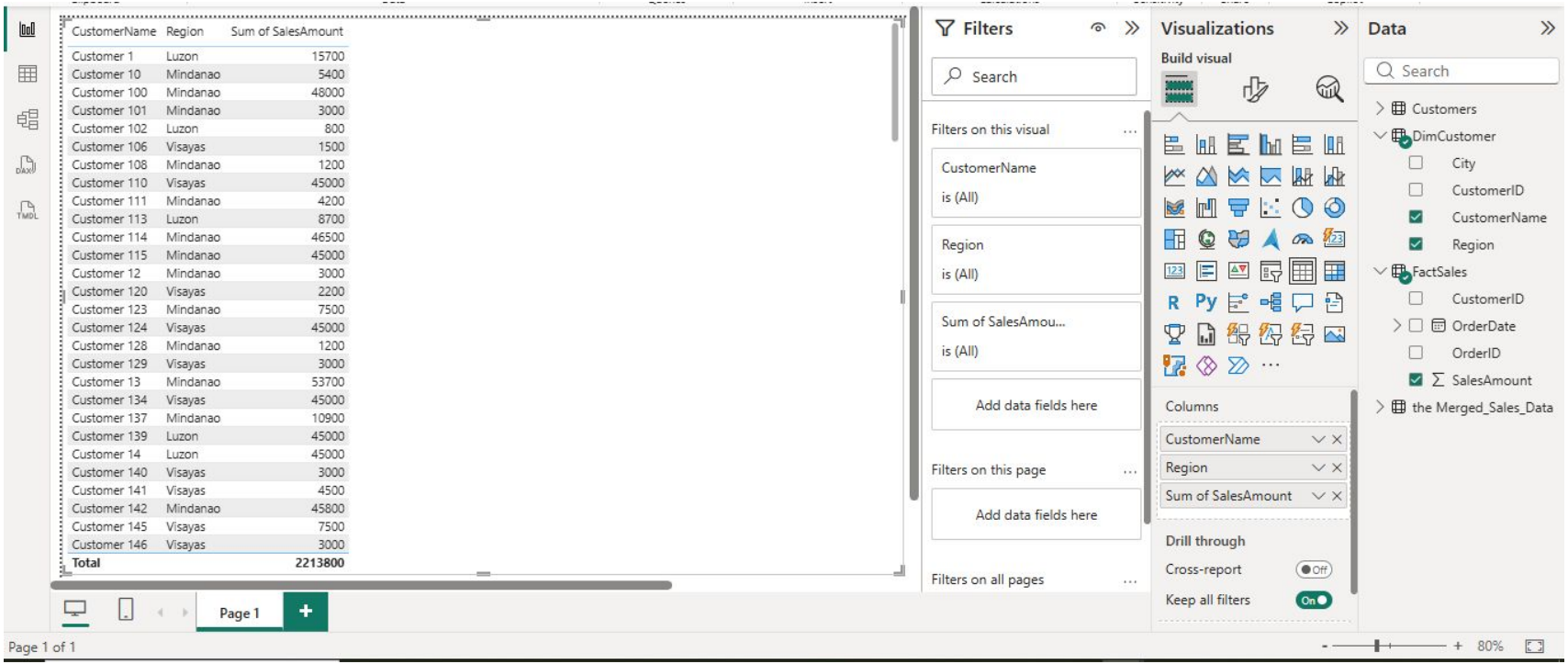
1. Why should relationships flow from dimension to fact?

   Because filters from dimension tables (such as customer or region) need to pass to the fact table so the data can be grouped and analyzed correctly.

2. What happens if the relationship is Many-to-Many?

   It can create unclear links between tables, which may result in duplicated counts and inaccurate report totals.

Part 5: Validate the Data Model



CustomerName	Region	Sum of SalesAmount
Customer 1	Luzon	15700
Customer 10	Mindanao	5400
Customer 100	Mindanao	48000
Customer 101	Mindanao	3000
Customer 102	Luzon	800
Customer 106	Visayas	1500
Customer 108	Mindanao	1200
Customer 110	Visayas	45000
Customer 111	Mindanao	4200
Customer 113	Luzon	8700
Customer 114	Mindanao	46500
Customer 115	Mindanao	45000
Customer 12	Mindanao	3000
Customer 120	Visayas	2200
Customer 123	Mindanao	7500
Customer 124	Visayas	45000
Customer 128	Mindanao	1200
Customer 129	Visayas	3000
Customer 13	Mindanao	53700
Customer 134	Visayas	45000
Customer 137	Mindanao	10900
Customer 139	Luzon	45000
Customer 14	Luzon	45000
Customer 140	Visayas	3000
Customer 141	Visayas	4500
Customer 142	Mindanao	45800
Customer 145	Visayas	7500
Customer 146	Visayas	3000
Total		**2213800**

Guide Questions

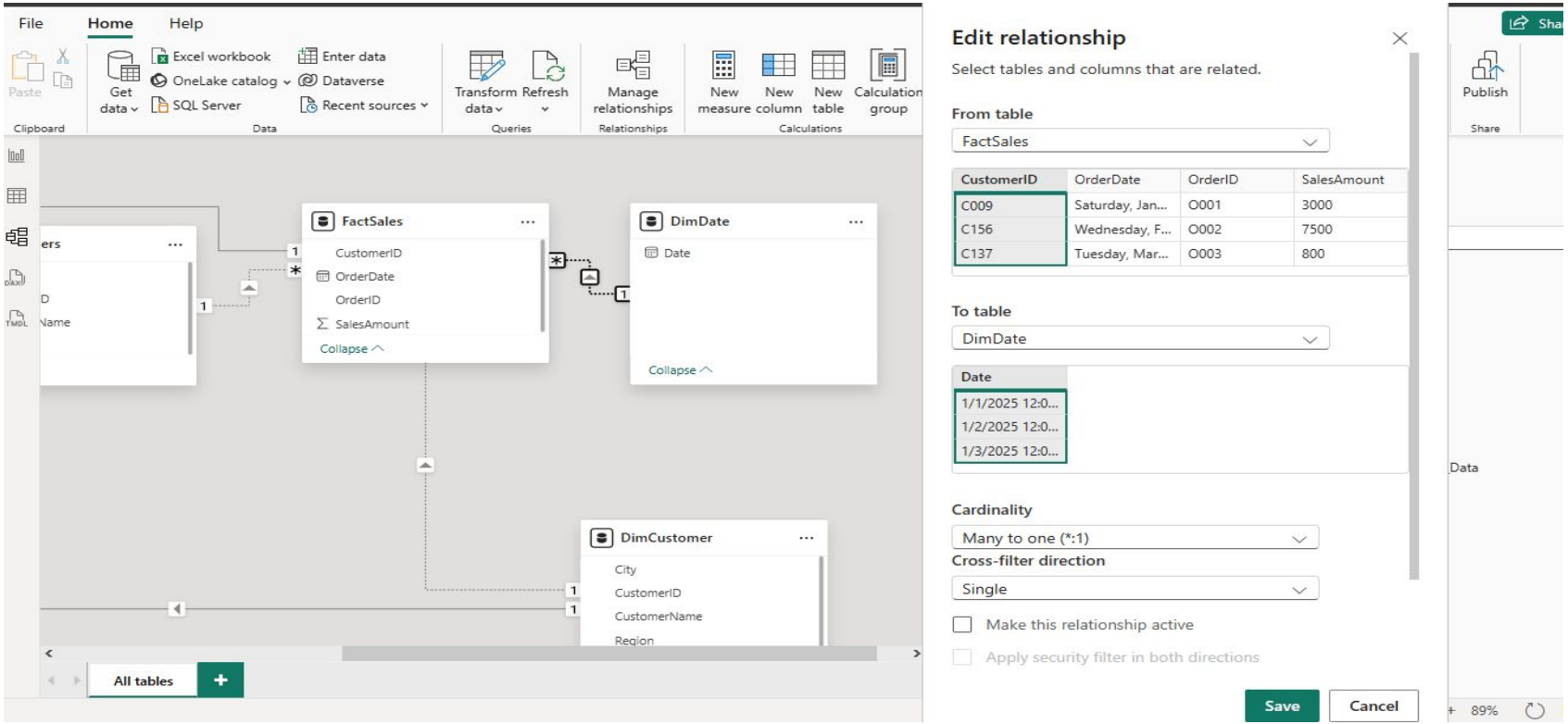
1. Did sales aggregate correctly per customer?

   Yes, the sales totals are correct because each customer shows their own total Sales Amount, which means the relationship between the tables is working.

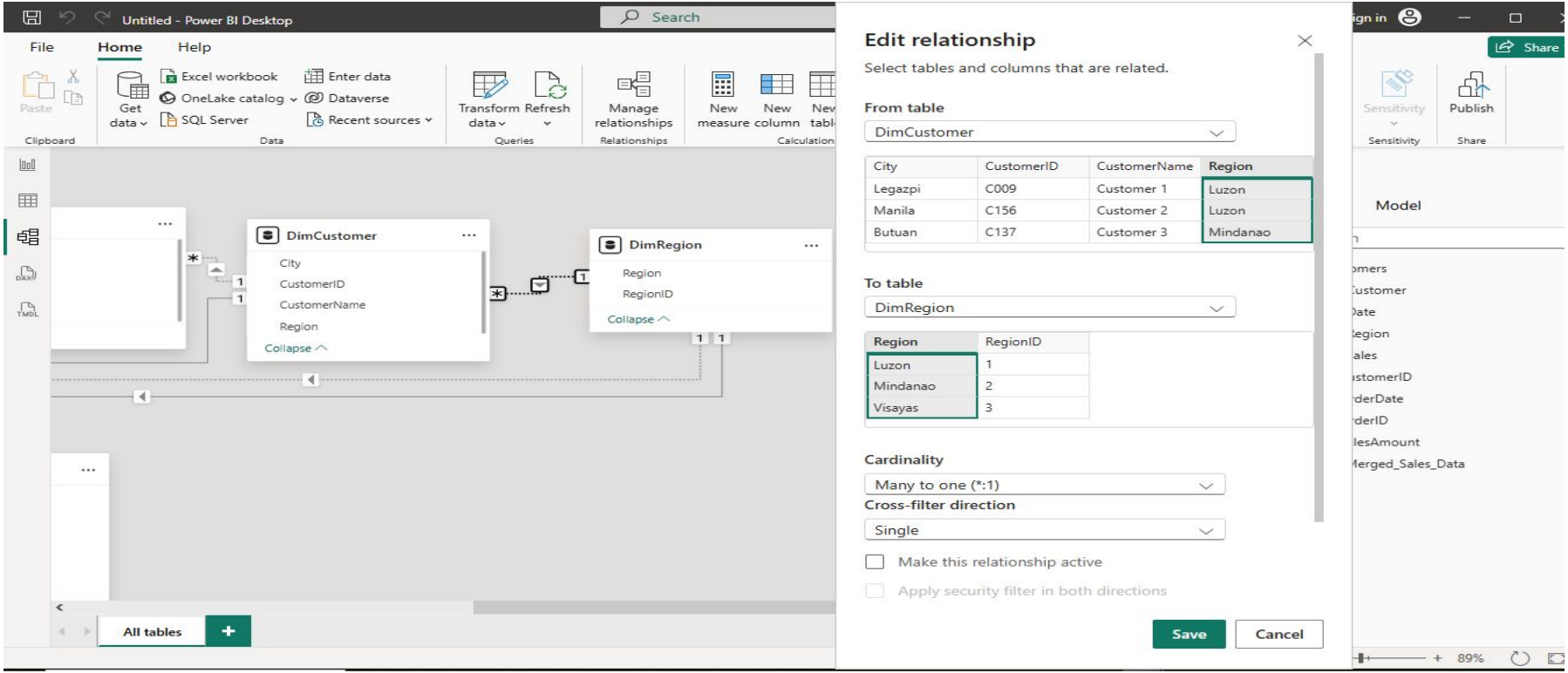
2. What would happen if relationships were missing?

   If there were no relationships, every customer would show the same total sales instead of their own amount, making the report incorrect.

Enhancement Activities

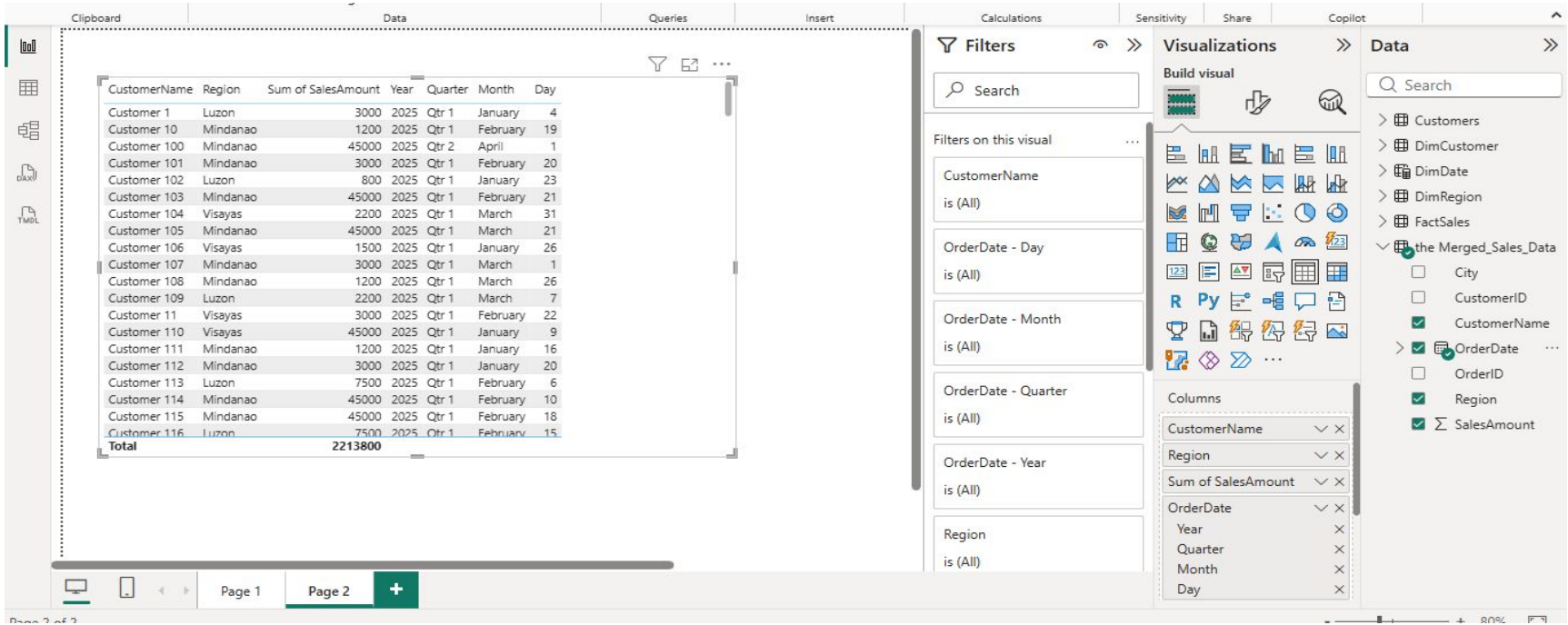


Enhancement 2: Create a DimRegion Table



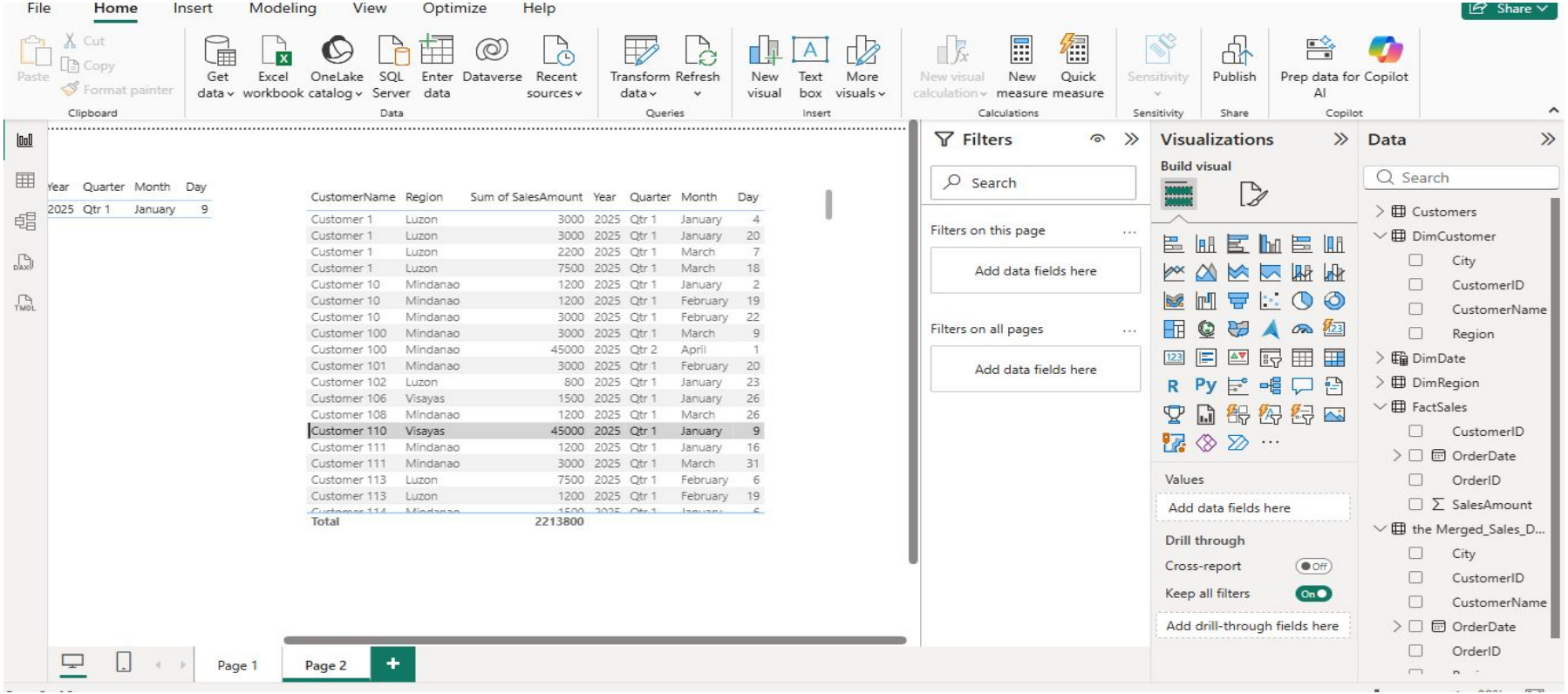
Enhancement 3: Compare Flat vs Star Schema Performance

- Create visuals using a flat table



CustomerName	Region	Sum of SalesAmount	Year	Quarter	Month	Day
Customer 1	Luzon	3000	2025	Qtr 1	January	4
Customer 10	Mindanao	1200	2025	Qtr 1	February	19
Customer 100	Mindanao	45000	2025	Qtr 2	April	1
Customer 101	Mindanao	3000	2025	Qtr 1	February	20
Customer 102	Luzon	800	2025	Qtr 1	January	23
Customer 103	Mindanao	45000	2025	Qtr 1	February	21
Customer 104	Visayas	2200	2025	Qtr 1	March	31
Customer 105	Mindanao	45000	2025	Qtr 1	March	21
Customer 106	Visayas	1500	2025	Qtr 1	January	26
Customer 107	Mindanao	3000	2025	Qtr 1	March	1
Customer 108	Mindanao	1200	2025	Qtr 1	March	26
Customer 109	Luzon	2200	2025	Qtr 1	March	7
Customer 11	Visayas	3000	2025	Qtr 1	February	22
Customer 110	Visayas	45000	2025	Qtr 1	January	9
Customer 111	Mindanao	1200	2025	Qtr 1	January	16
Customer 112	Mindanao	3000	2025	Qtr 1	January	20
Customer 113	Luzon	7500	2025	Qtr 1	February	6
Customer 114	Mindanao	45000	2025	Qtr 1	February	10
Customer 115	Mindanao	45000	2025	Qtr 1	February	18
Customer 116	Luzon	7500	2025	Qtr 1	February	15
Total		2213800				

- Create visuals using a star schema

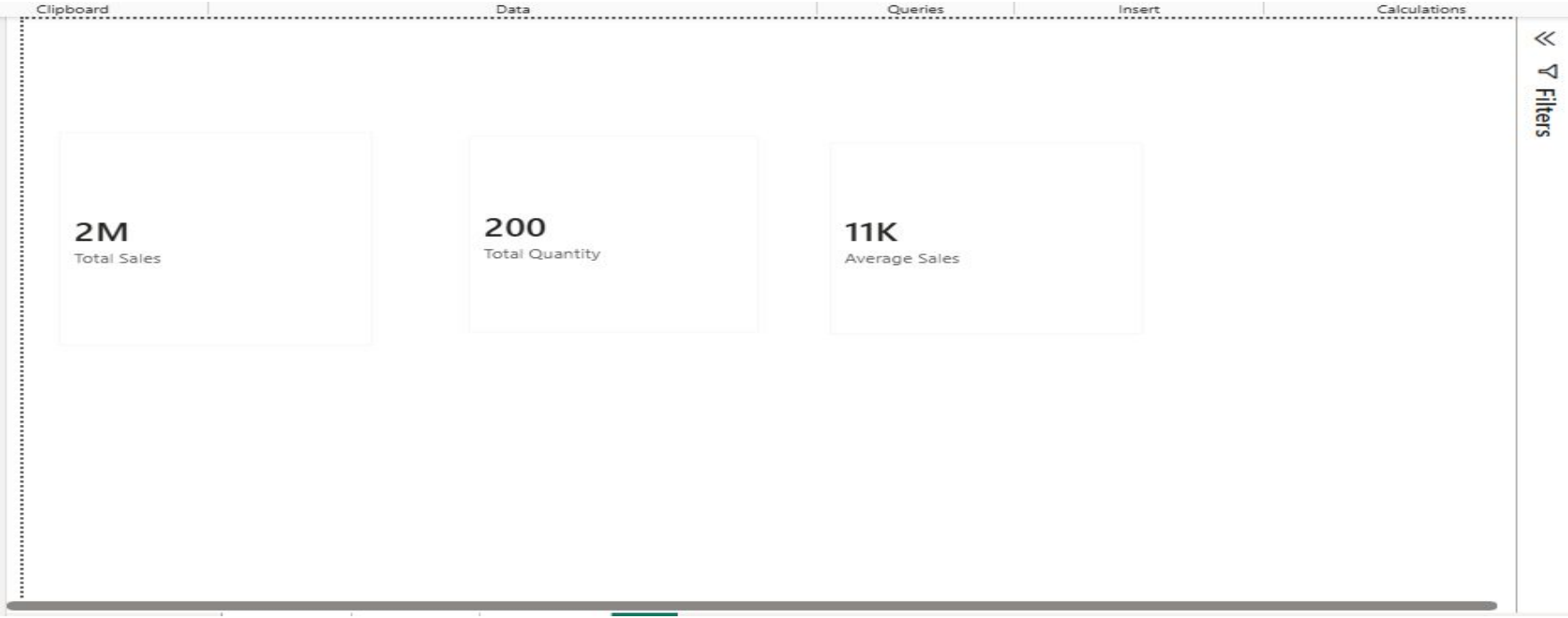


- Observe the performance difference

The star schema visual filters faster and more accurately because data is organized into separate dimension and fact tables, reducing redundancy compared to the flat table.

Introduction to DAX

Create Measures



Guide Questions

1. What is a DAX measure?

A DAX measure is a formula used to calculate values in a report. It updates automatically depending on filters or slicers.

2. Difference between SUM and AVERAGE?

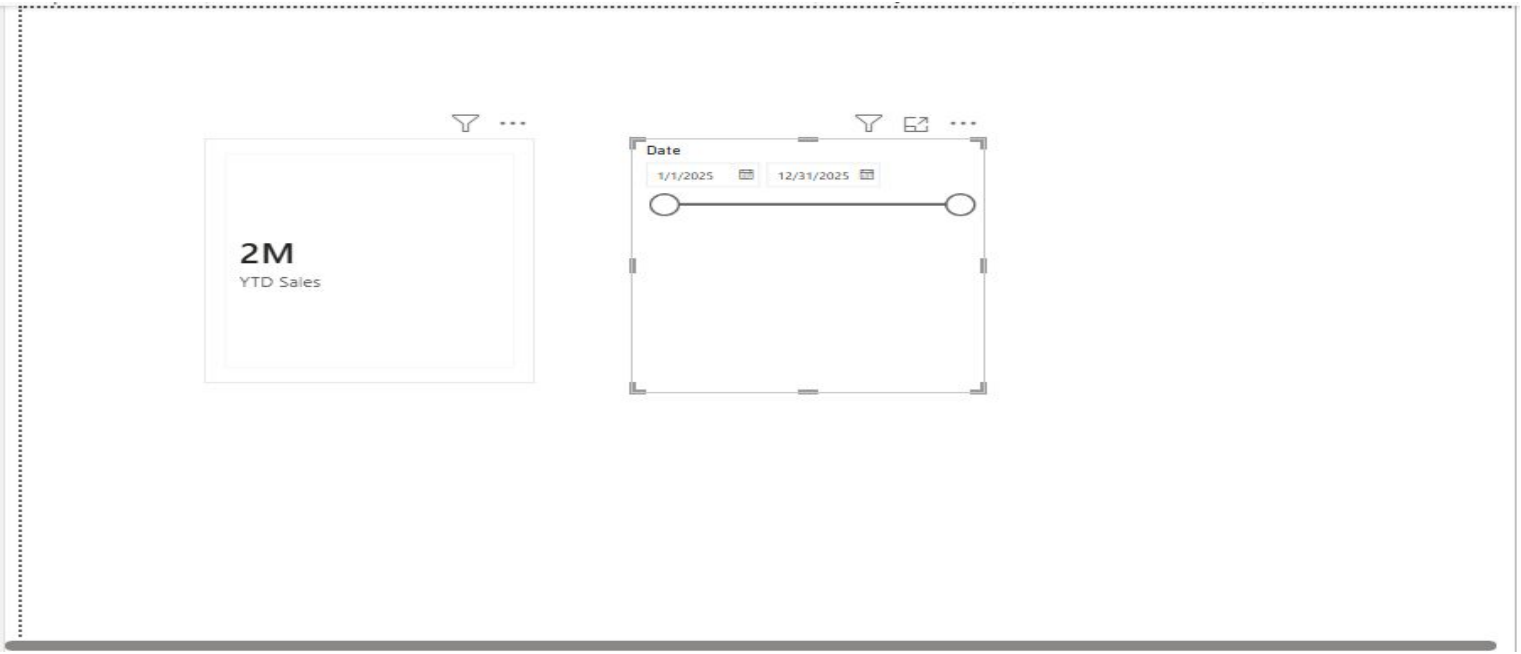
SUM adds all numbers to get the total, while AVERAGE finds the middle value by dividing the total by the number of items.

3. Why use measures instead of calculated columns?

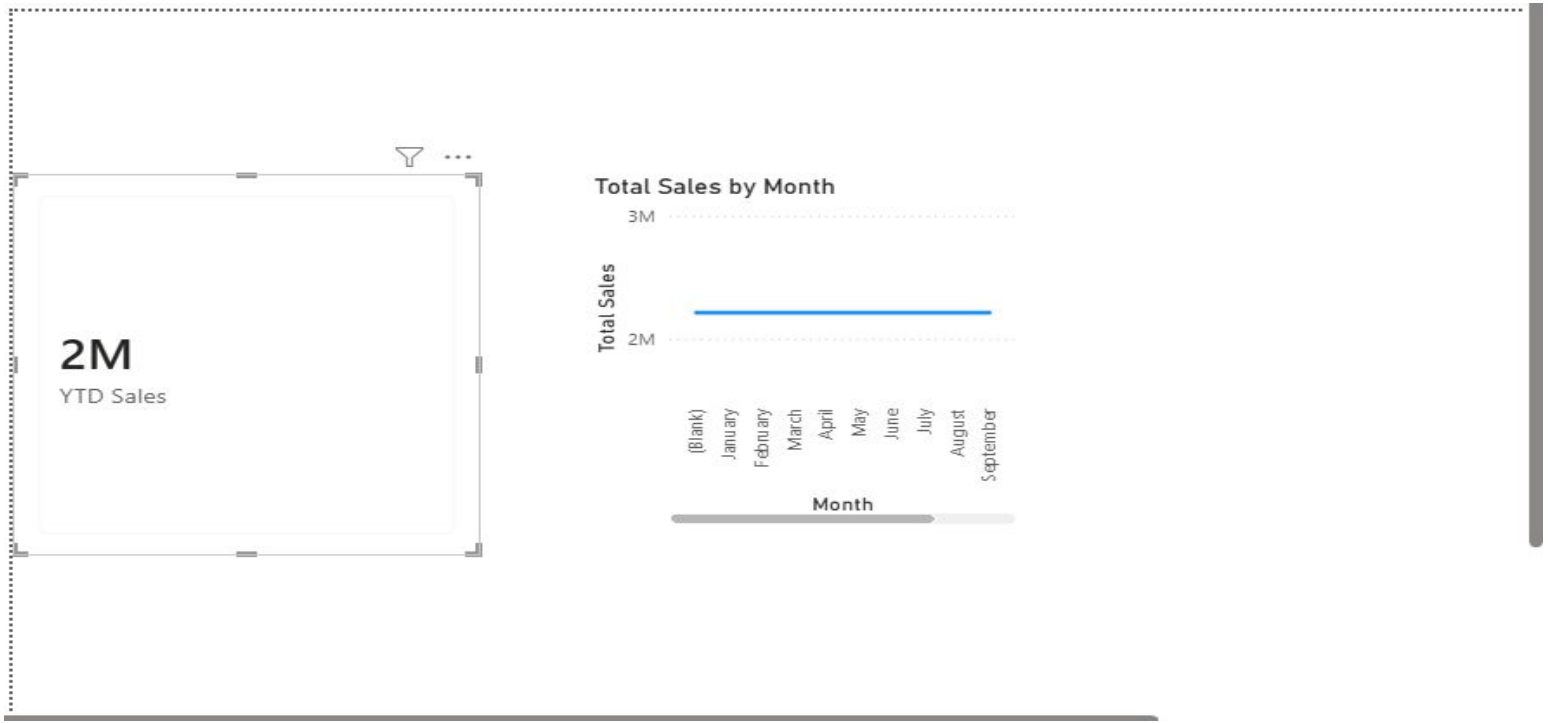
Measures calculate results only when needed and adjust to filters, making them more efficient. Calculated columns are stored in the table and use more memory.

Enhancement Activities DAX Functions

1. Year-to-Date Sales



2. Monthly Sales Growth



Guide Questions

1. What is time intelligence in DAX?

Time intelligence uses DAX functions to calculate values over time, like year-to-date, month-to-month, or previous periods.

2. Why is a date table required?

A date table is needed because functions like TOTALYTD or PREVIOUSMONTH need a complete sequence of dates to work correctly.

3. How does PREVIOUSMONTH help analyze trends?

PREVIOUSMONTH gets the total from the month before, letting you compare it to the current month to see if sales are rising or falling.

4. What insights can YTD Sales provide?

YTD Sales shows the total from the start of the year up to now, helping track progress toward yearly sales goals.